Assignment 4

① $N = \left(\frac{1.96}{0.2}\right)^2 (0.1)(0.9) = 864.36$

$= 865$ simulations.

② (done in Jupyter)

③ ↗

④ $SE = \sqrt{\frac{0.305(1-0.305)}{N}}$ $N = 16(8 \times 365) = 46720$

$= 0.0021$

$\varepsilon = 1.96(SE) = 0.0042$

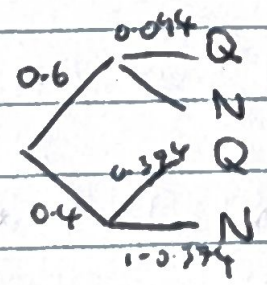
Confidence $= 0.305 \pm 0.004$

⑤ Jupyter

⑥ seconds, 110 in steps of 5
107 ↗

⑦

⑧



$0.6(0.094) + 0.4(0.374) = 0.206$

(9) The program ~~determines~~ randomly selects whether roadwork is happening or not with 40% weighting, ~~isted~~ instead of a fixed 40% of the time.

~~(10) Jupyter~~

~~(11) [illegible]~~

(10) – (13) Jupyter

(14) separate page.

```
[1]: import numpy as np

     np.random.seed(3500)

[2]: A = []
     B = []

     success = 0

     nsim = 5000

[3]: for i in range(nsim):
         A.append(np.random.random())
         B.append(np.random.random())

         if abs(A[i] - B[i]) < 0.2:
             success += 1

     print("Probability that distance is less than 0.2 = ", success / nsim)

     Probability that distance is less than 0.2 =  0.353
```